Emaan Heidari

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Education

University of Southern California - GPA: 3.97/4.00 Expected Dec 2027
Bachelor of Science in Computer Engineering and Computer Science *Los Angeles, CA*

- Coursework: Parallel Computing, Algorithms, Data Structures, Machine Learning, Operating Systems, Distributed Systems, Computer Systems, Embedded Systems, Digital Design, Networking, Linear Algebra, Probability/Statistics

Experience

Tesla May 2026 – Present
Software Engineering Intern *Palo Alto, CA*

- Architected and deployed an on-device CAN broker in Rust, reducing p99 end-to-end latency by 98% to 1.3 ms and peak memory footprint by 96% over a legacy C++ service, with gRPC APIs for local Go clients across 1M+ devices
- Optimized a bus multiplexer Linux daemon in Rust, increasing throughput by 21% through perf profiling, heap analysis, batched Protobuf IPC, and buffer reuse; deployed on 300K+ Tesla Powerwalls globally

USC Liquid Propulsion Lab Sep 2025 – Present
Software Engineer *Los Angeles, CA*

- Led the design of a C++20 telemetry system for liquid rocket engines, processing 80K+ samples/sec from 38 sensors
- Implemented a low-latency control loop to automate valve-abort commands within 4 ms of a faulting sensor reading
- Built a telemetry replay and fault-injection SIL framework in Python using pytest, InfluxDB, and Docker

SiFly May 2025 – Sep 2025
Software Engineering Intern *Santa Clara, CA*

- Improved obstacle detection accuracy from 50% to 82% at long range while maintaining 92% precision by fusing depth data from PyTorch and OpenCV models with NumPy across simulated drone environments
- Built a real-time ROS 2 perception pipeline hitting 40 FPS at 35 ms on an RTX 4090, profiled with Nsight Systems
- Extended PX4's C++17 flight stack with new failsafe modes for autonomous mission resumption after control loss

Tandem Launch May 2024 – Aug 2024
Software Engineering Intern *Montreal, Canada*

- Engineered a CNN inference runtime in TensorFlow and NumPy for an analog AI accelerator, collaborating with hardware researchers to raise accuracy from 68% to 93% via hyperparameter search and quantization-aware training
- Cut hardware-in-the-loop firmware testing cycles 90% by designing a Docker-based CI infrastructure on Linux

Projects

Efficient Decode Attention | *C++, CUDA, CMake, Nsight Compute, Google Test + Benchmark* [PDF] [Code]

- Implemented a flash-decode CUDA kernel eliminating the softmax bottleneck, achieving 28x throughput over a standard attention kernel (477 GiB/s vs. 17 GiB/s) on an NVIDIA A40
- Designed a paged KV cache manager in C++20 eliminating fragmentation via fixed-size page allocation, supporting 4-8x more concurrent sequences at a constant 1.8x per-token throughput cost

MurmurMatch | *Python, GCP, PostgreSQL, FastAPI, NodeJS, React, Nginx, Google OAuth, Stripe* [Website]

- Built and scaled a college social platform to 56,000+ users used by 1/2 of Dartmouth, 1/3 of Brown, 1/3 of Stanford, and 1/4 of Princeton students, converting 500+ users to a $7.99/week subscription
- Architected load-balanced GCP infrastructure with autoscaling Managed Instance Groups to support demand

FPGA Neural Network Inference Accelerator | *Verilog, Vivado, Python, PyTorch* [PDF] [Code]

- Built a pipelined FPGA neural inference accelerator in Verilog with 8.95μs latency and 3.43μJ energy per inference
- Implemented fixed-point deterministic inference with LUT activations and QAT, reaching 95.1% accuracy

Skills

Languages: C++, Python, Rust, Go, C, Java, SQL, Bash, TypeScript, JavaScript, Verilog
Technologies: CUDA, PyTorch, TensorFlow, Nsight Compute, vLLM, TensorRT, Git, Linux, CMake, perf, gdb, Valgrind, OpenMP, MPI, Pthreads, Spark, NumPy, Docker, AWS, GCP, FastAPI, Zephyr, gRPC, Protobuf, PostgreSQL